

Claudia Vanessa Hermosillo Diaz

COMPUTER SCIENCE AND TECHNOLOGY UNDERGRADUATE — RESEARCH & COMPUTATIONAL SYSTEMS

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PROFILE

Computer Science and Technology undergraduate at Tecnológico de Monterrey with hands-on experience contributing to computational research projects in biomolecular simulation and autonomous robotics through the Next Gen Scientist program. Comfortable working inside existing research codebases in C, CUDA, and Python, and translating unfamiliar computational methods into working GPU-accelerated implementations under the guidance of a lead researcher. Additional technical background in cybersecurity, networking, and iOS/geospatial software development. Combines fast independent learning with experience leading and mentoring technical teams.

EDUCATION

Computer Science and Technology | 2024 – Present

Instituto Tecnológico y de Estudios Superiores de Monterrey, Campus Toluca

RESEARCH EXPERIENCE

Computational Biomolecular Research — Next Gen Scientist Program, Tec de Monterrey

Research Member (Computational/Technical Support) April 2025 - June 2026

- Contributed to an ongoing computational biomolecular research project modeling RNA and biomolecular interactions through Brownian dynamics simulation.
- Studied and debugged an existing C-based Brownian dynamics simulation to understand the team's computational methodology, then translated selected components toward CUDA under the guidance of the lead researcher to enable GPU acceleration.
- Worked directly with force-calculation components of the simulation, including bonding, flexibility, torsion, and Watson-Crick interaction terms.
- Presented project progress at the program's inaugural research encounter.

Autonomous Robotics Research — Next Gen Scientist Program, Tec de Monterrey

Co-Founding Research Member (Ongoing)

- Co-founded an autonomous robotics research project with a student team, proposing the initial project focus and scope under the supervision of a faculty advisor.
- Contributing to ongoing development of navigation and hardware-level optimization on an NVIDIA Jetson Orin Nano platform.

TECHNICAL PROJECTS

NAVIA — iOS Personal Safety Application (Swift, Mapbox SDK) | In Development

- Building an iOS application to support urban mobility safety, using Swift, Combine, and an MVVM architecture.
- Designing real-time, geospatial routing logic with the Mapbox SDK to calculate lower-risk urban routes, including geofencing and safety-state logic.
- Implementing user-validation and anti-tamper safeguards to help protect account integrity.

AI for Mexican Sign Language (LSM) | In Development

- Developing a vision-based model for real-time translation of Mexican Sign Language.
- Using MediaPipe and LSTM with TensorFlow

TECHNICAL EXPERIENCE & LEADERSHIP

FIRST Robotics Competition (FRC) & Tech Challenge (FTC) | 2022 – 2026

Robot Programmer (2022–2024); Robot Inspector, Referee & Team Leader (2022–2026)

- Built an autonomous scoring routine for the FTC robot using RoadRunner for odometry-based path following, prototyping and tuning trajectories in MeepMeep before deployment.

- Trained a custom TensorFlow Lite object-detection model to identify a scoring game element via camera in real time, triggering the autonomous scoring sequence.
- Implemented PID and feedforward (FF) control for robot mechanisms to execute autonomous movements and scoring actions smoothly.
- Inspected electromechanical hardware and embedded Java control software against competition technical rules, identifying non-compliant systems and working with teams on fixes under event deadlines.
- Monitored live matches to enforce competition rules and resolve real-time disputes during execution.
- Mentored new team members in Java-based control systems programming and debugging.

Global Leader & STEM Outreach

- Eugenio Garza Sada Global Leadership Program, 5th Generation: completed a leadership track connecting technical projects to UN Sustainable Development Goals (SDG 5, 11, 16) focused on urban safety; graduated as Silver Dandelion (top 20% of the generation).
- Eugenio Garza Sada Global Leadership Program, ILO Mentor, 6th Generation: mentoring incoming students in the Individual Leadership Opportunity track.
- STEM Outreach: taught coding and engineering fundamentals to over 30 middle school girls and local students to promote gender equality in STEM.
- Estudiante Embajadora de Google 2026 — Inaugural Generation: selected into the first cohort of a national educational community connecting university students across Mexico to exchange practical experience with Google Gemini and AI tools, as part of a digital-training and leadership program aimed at developing local AI reference points.

TECHNICAL SKILLS

Programming & Computation: Python (scripting & AI), C, C++, CUDA, Java (Oracle Certified), Swift.

Robotics & Controls: RoadRunner, MeepMeep, odometry-based path planning, PID/feedforward control, TensorFlow Lite object detection.

Platforms & Tools: Linux (Ubuntu/Jetson), NVIDIA Jetson SDK, Git/GitHub, VS Code.

Cybersecurity & Networking: Network fundamentals, secure routing protocols, system risk assessment, technical compliance auditing (Cisco CCNA 1).

AI Tools: Google Gemini / Applied AI (Estudiante Embajadora de Google 2026).

Soft Skills: Leadership, mentorship, technical problem-solving, cross-team collaboration.

Languages: Spanish (Native), English (C1), German (Basic), French (Basic).

CERTIFICATIONS

- Cisco Networking Academy — CCNA 1: Introduction to Networks (Completed Jun 2026).
- Cisco Networking Academy — CCNA 2: Switching, Routing, and Wireless Essentials (In Progress).
- Cisco Networking Academy — Introduction to Cybersecurity (Completed Jul 2026).
- Cisco Networking Academy — Cybersecurity Fundamentals (In Progress).
- Oracle — Java Fundamentals, Java Programming (Completed Nov 2025, Dec 2025).
- Google — Init.g: Software Engineering Foundations (Completed Oct 2025).